A Feasibility-Aware Quantum-Classical Benchmark for Occlusion-Aware UAV Trajectory Optimization as a HUBO

***Abstract*—We study discrete UAV trajectory optimization with obstacle avoidance *and* line-of-sight occlusion penalties, formulated as a Higher-Order Unconstrained Binary Optimization (HUBO) problem derived from the multi-convex MPC tracking problem of Masnavi *et al.* [1]. Unlike recent quantum UAV work that targets obstacle avoidance [2] or routing/logistics [3], our formulation explicitly encodes target-visibility preservation, and uses a genuinely cubic proximity term that resists QUBO reduction. We make three contributions: (i) we frame occlusion-aware UAV trajectory optimization as a quantum-friendly discrete benchmark; (ii) we propose a *feasibility-aware* benchmarking methodology that separates hard-constraint satisfaction from soft-cost quality, uses independent constraint verification rather than energy thresholds, matches sampling budgets across solvers, and characterizes constraint exploitation through penalty sweeps; and (iii) we analyze the higher-order (HUBO) structure against the cost of QUBO flattening. On the full 960-variable instance, SA reaches 39 % lower soft cost than D-Wave neal but is 187× slower; neal achieves only 22 % true feasibility, with an exploitation curve rising from $0\,\%$ at λ=50 to $60\,\%$ at λ=2000. QAOA reproduces the optimum on a tiny instance but does not scale.**

***Index Terms*—HUBO, QUBO, quantum annealing, QAOA, occlusion-aware planning, UAV trajectory optimization, feasibility, benchmarking, D-Wave, Qiskit.**

I. Introduction

Recent work has applied quantum and quantum-inspired methods to UAV problems from two directions. QUAV [2] addresses single-UAV *obstacle-avoidance* path planning with QAOA, validated on small/path-segmented instances against A* and RRT. Q4DR [3] addresses drone *routing* through a hybrid decomposition — QAOA clustering followed by D-Wave annealing for the routing stage. Neither line of work addresses *line-of-sight preservation* or *occlusion-aware tracking*, where the UAV must not only avoid obstacles but also keep a target visible.

We position our work in the gap between these: discrete UAV trajectory optimization in which obstacle cells *and* target occlusion both incur penalties. Building on Masnavi *et al.* [1], we reformulate the problem as a Higher-Order Unconstrained Binary Optimization (HUBO) [4], [6] amenable to annealing and, after quadratization, to QAOA [5].

Our contributions are threefold:

1) **Occlusion-aware formulation.** We encode target-visibility preservation directly in the cost function, distinguishing this work from obstacle-only [2] and routing-only [3] quantum UAV studies.
2) **Feasibility-aware benchmarking.** We propose a benchmarking methodology that separates hard feasibility from soft quality, uses independent constraint checks, matches sampling budgets, and sweeps penalty weights — exposing failure modes that raw-energy comparisons hide.
3) **Higher-order structure analysis.** We show why occlusion and sustained-proximity penalties are naturally cubic, and what is lost when the problem is flattened to QUBO.

II. Occlusion-Aware HUBO Formulation

Let V be the set of grid cells and T the planning horizon. The decision variable $x[t,v] \in \{0,1\}$ is 1 iff the UAV occupies cell v at time t. The Hamiltonian has seven terms:

$$H(x) = H_{\text{uniq}} + H_{\text{start}} + H_{\text{move}} + H_{\text{obs}} + H_{\text{occ}} + H_{\text{prox}} + H_{\text{term}}. \tag{1}$$

H_{uniq}, H_{start}, H_{move} enforce kinematic constraints quadratically (one cell per step, fixed start, 4-connected motion); H_{obs} forbids obstacle cells.

Occlusion penalty. H_{occ} is the application-defining term. For each cell u, we trace the Bresenham line-of-sight to the target and count intervening obstacle cells:

$$H_{\text{occ}} = \lambda_{\text{occ}} \sum_{t} \sum_{u \in V} x[t,u] \,\big|\, \text{LOS}(u, \text{target}) \cap O \,\big|, \tag{2}$$

where O is the obstacle set. This penalizes positions from which the target is occluded (Fig. 1). For a static target it is linear; for a *moving* target it becomes a joint product of UAV and target occupancy — intrinsically higher-order.

Genuinely-cubic proximity. The sustained-proximity term penalizes lingering in an obstacle-adjacent buffer cell v_b across three consecutive steps:

$$H_{\text{prox}} = \lambda_{\text{prox}} \sum_{v_b \in B} \sum_{t=1}^{T-1} x[t-1, v_b]\, x[t, v_b]\, x[t+1, v_b]. \tag{3}$$

Each summand is a product of three distinct variables; transient passage is permitted, only lingering is penalized.

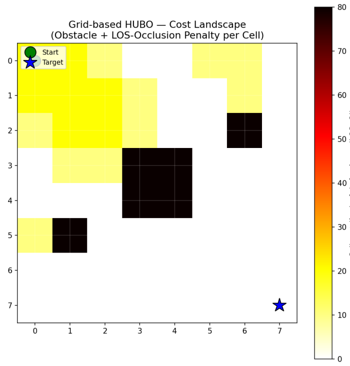

Fig. 1: Per-cell cost: obstacle penalty plus line-of-sight occlusion to the target. Dark cells are obstacles; yellow cells incur occlusion penalty because the target is hidden behind an obstacle from there.

TABLE I: Feasibility-aware benchmarking: each safeguard prevents a specific failure mode.

Safeguard	Failure mode it prevents
Energy decomposition (hard penalty vs. soft cost)	Confusing constraint satisfaction with trajectory quality
Independent constraint check (decode + verify)	False feasibility from energy-threshold heuristics
Matched sampling budget (equal reads/runs)	Unfair advantage to whichever solver gets more samples
Penalty-weight sweep	Reporting one λ that hides the exploitation regime

Why HUBO, not QUBO. Eq. (3) cannot be made quadratic without Rosenberg auxiliaries — one per cubic term [4]. For our instance this adds ~3,172 ancillas (~30 % more variables) with dense couplings, degrading the sparsity annealers exploit. The full instance has **960 variables**, **87,684 monomials** (3,172 cubic).

III. Feasibility-Aware Benchmark Methodology

A central contribution of this work is the benchmarking methodology itself. Raw HUBO energy is a misleading comparison metric because penalty-based solvers can lower energy by *violating* soft constraints. We adopt four safeguards (Table I).

Since Qiskit's state-vector simulator scales as 2^n and cannot run at n=960, we use two tiers: **Tier 1** (full 8×8, 960 vars: SA vs. neal) and **Tier 2** (tiny 2×2, 8 vars: all three solvers). We use 50 matched samples per solver, SA at 500 sweeps, and hardened weights λ=1000.

IV. Results

A. Tier 1 — Quality, Speed, Feasibility

On 50 matched samples, SA attains 39 % lower soft cost than neal (**303** vs. 496) but is 187× slower per sample (47.8 s vs. **0.26 s**). SA is feasible by construction (**100 %**); neal reaches only 22 % true feasibility. SA reaches the target in 14 % of runs, neal in 8 %. Best total energies are −15,697 (SA) and −15,504 (neal). Notably, SA's *mean* soft cost (~495) matches neal's *best* feasible soft cost (496): neal needs many reads to occasionally equal SA's typical quality.

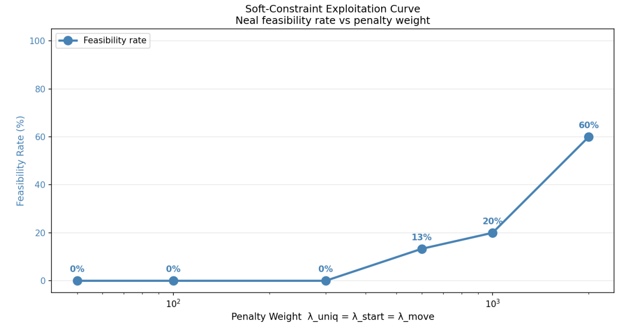

Fig. 2: Constraint-exploitation curve: neal feasibility vs. penalty weight λ. General-purpose annealers need weights an order of magnitude above the objective scale to respect one-hot HUBO structure.

B. The Constraint-Exploitation Curve

Sweeping λ (Fig. 2) reveals the failure mode the methodology targets. At $\lambda \leq 300$, neal achieves 0 % feasibility — every low-energy sample violates constraints. Feasibility rises to 13 % at λ=600, 20 % at λ=1000, and 60 % at λ=2000. A single-λ report would mis-state neal's capability; the sweep makes the exploitation regime explicit. In the best feasible runs, SA reaches $(7, 7)$ while neal stops at $(5, 4)$; both avoid obstacles and high-occlusion cells.

C. Tier 2 — Three-Way Comparison

On the 8-variable instance all three solvers reach the same optimum E^*= − 118.17, confirming the QAOA implementation (QuadraticProgram → QAOA, SPSA, reps=1). QAOA takes 55 s vs. 0.02 s (neal) and 0.14 s (SA) — a 2,750× overhead from state-vector simulation that grows exponentially with qubit count.

V. Discussion and Conclusion

We position this work as a *feasibility-aware quantum-classical benchmark for occlusion-aware discrete UAV trajectory optimization*, not a claim of quantum advantage. Occlusion-awareness distinguishes the formulation from obstacle-only [2] and routing [3] quantum UAV work, and the visibility relation becomes intrinsically cubic once the target moves. Feasibility-aware benchmarking proves necessary: decomposition, independent checks, and penalty sweeps reveal that 78 % of neal's hardened-weight samples still violate constraints — a fact raw energy hides. Future work: a moving target (fully cubic occlusion), real D-Wave hardware, and comparison with the multi-convex MPC baseline.

References

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